

Relational Calculus (Logic)



$$\{p.prod_name \mid Product(p) \wedge p.prod_price > 20 \wedge p.prod_manufacturer = \text{‘GizmoWorks’}\}$$

Product

prod_name	prod_price	prod_manufacturer
Gizmo	$19.99	GizmoWorks
Powergizmo	$39.99	GizmoWorks
Widget	$19.99	WidgetsRUs
HyperWidget	$203.99	Hyper
VaporWare	NULL	NULL

$$\{p.prod_price \times 0.0825 \mid \exists m(Product(p) \wedge Manufacturer(m) \wedge p.prod_manufacturer = m.man_name \wedge m.man_city = \text{‘Houston’} \wedge m.man_state = \text{‘TX’})\}$$

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Manufacturer

man_name	man_address	man_city	man_state
GizmoWorks	123 Gizmo St.	Houston	TX
WidgetsRUs	20 Main St.	New York	NY
Hyper	1 Mission Dr.	San Francisco	CA



$$\{m.man_name \mid Manufacturer(m) \wedge \neg\exists p(Product(p) \wedge p.prod_manufacturer = m.man_name)\}$$

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Notational Details

- Set-former notation: $\{x.\, a, f(y.\, b), \ldots | \varphi\}$

- $Product(p)$ also written as $p \in Product$.

Quantification

$$\{p.prod_price \times 0.0825 \mid \exists m(Product(p) \wedge Manufacturer(m) \wedge p.prod_manufacturer = m.man_name \wedge m.man_city = \text{‘Houston’} \wedge m.man_state = \text{‘TX’})\}$$

$$\{m.man_name \mid Manufacturer(m) \wedge \neg\exists p(Product(p) \wedge p.prod_manufacturer = m.man_name)\}$$